1 · What PRAMAAN-AI does

- **PRAMAAN-AI is a decision-support layer for AICTE applicatio reads submitted evidence, extracts typed claims, runs five ind verification engines, and produces a Trust Score with linked i evidence.**
- **GREEN means fast-track desk review, AMBER means focused RED means priority EVC inspection. These labels never appro institution.**
- **The system is intentionally deterministic where possible. Opti for perception and claim suggestions; APH rules, hard gates a decisions remain authoritative.**

2 · End-to-end data flow

- **Step 1 — A reviewer logs in with a role. Evaluator reviews and Official records the final decision; Admin governs; Auditor ve**
- **Step 2 — Documents are classified and text is extracted. PDF active in the MVP; OCR and layout models are replaceable ada**
- **Step 3 — The Claim Graph stores a value, confidence, source and bounding-region reference.**
- **Step 4 — PARSE, PROVE, MATCH, VERIFY and SEE evaluate independently. A failure of one engine does not silently disap**
- **Step 5 — Weighted pillars and hard gates produce the score. exact evidence. The official accepts, overrides with a reason, decision.**
- **Step 6 — Every ingest, engine run, flag action, decision and v appended to a SHA-256 hash chain.**

3 · The five engines

- **PARSE checks mandatory evidence, expiry and basic claim ar the versioned APH Rule Graph.**
- **PROVE computes block-wise ELA and exposes forensic signa is ‘inconsistent with an untampered original; inspect here’, ne forgery verdict.**
- **MATCH compares historical metrics and salted faculty identifi graph key is evidence for inspection, not a public identity disc**
- **VERIFY ranks source strength: signed issuer source, DSC, QR and registry receipt. An unreachable issuer becomes a manua an invalid submission.**
- **SEE checks geofence, image reuse and future object-count se is a precise EVC agenda rather than a generic visit request.**

4 · Trust Score and governance

- **Pillars: completeness 20%, authenticity 25%, consistency 20% verification 20%, infrastructure plausibility 15%.**
- **Hard gates: an unresolved HIGH flag caps the score at 58; a c mandatory issuer-verification failure caps it at 40.**
- **The score is a prioritisation aid. The dashboard enforces role- and requires reasons for overrides and negative decisions.**
- **Impact values in the project are design targets until validated Seeded files and prototype outputs are synthetic and labelled**

5 · AI provider policy — no leaked keys

- There are no legitimate ‘open-source API keys’ to copy from th keys are usually stolen, revoked, rate-limited or unsafe to use in JavaScript, Git, a PDF, or a screenshot.
- The default is rules-only mode: AI_PROVIDER=disabled. The a works and the regulatory checks remain auditable.
- For a no-key local model, install Ollama and run a model on th machine. Local Ollama at http://localhost:11434 does not requ authentication.
- For an online provider, OpenRouter is supported through AI_PROVIDER=openrouter, OPENROUTER_API_KEY and OPENROUTER_MODEL=openrouter/free. A generic OpenAI-co endpoint is supported through PRAMAAN_AI_BASE_URL, PRAMAAN_AI_MODEL and PRAMAAN_AI_API_KEY. The statu reports only whether a key is present, never its value.
- The optional provider can suggest claims, but suggestions ar advisory_only and are not used to auto-approve, auto-reject o gates.

6 · Local Ollama setup (no API key)

- Install Ollama from https://ollama.com/download and start the
- Download a suitable local model, for example: ollama pull gem
a model that fits the available RAM/GPU and the institution's c
- From pramaan-app, set: export AI_PROVIDER=ollama; export
OLLAMA_BASE_URL=http://localhost:11434; export
OLLAMA_MODEL=gemma3:4b.
- Start the app with ./run.sh, log in, and call GET /api/ai/status th
dashboard session. 'available: true' means the local endpoint
- If Ollama is stopped or slow, the request falls back to determi
model outage never changes a compliance result.

7 · Hosted AI and official integrations

- Create credentials only in the provider’s official console. Stor Environment Variables or a secret manager; use separate Dev Preview and Production values. OpenRouter free models are r still require your own key.
- DigiLocker/API Setu is not an open anonymous API. It require registration, consent and issued credentials such as client ID, product instance ID depending on the integration path.
- Use official sandbox credentials and signed receipts. A copie uploaded to DigiLocker must not be treated as issuer-authenti because it is hosted there.
- For PaddleOCR, local deployment is open source. Hosted Pad services require their own access token. Add that integration server-side adapter and preserve confidence/routing metadata

8 · Run, test and demonstrate

- **Install: cd pramaan-app; python3 -m venv .venv; . .venv/bin/ac -r requirements.txt.**
- **Run: ./run.sh, then open http://localhost:8000. Demo passwor senior, admin and auditor is demo123.**
- **Walkthrough: log in as evaluator → open RED file → inspect P and page evidence → accept or override with reason → switch Official → record a human decision → verify audit chain.**
- **Offline engine: python3 pramaan-ai/engine/pramaan_engine.p clean/tampered ELA risk, the 52/100 demo score and audit val**
- **Checks: compile Python, run the API flow, verify Evaluator ap 403, verify override without reason returns 400, and verify the VALID.**

9 · Deployment and production hardening

- The Vercel demo uses a Python function and ephemeral /tmp storage. It is suitable for a demo, not durable production reco
- Production must move persistence to PostgreSQL, evidence t object storage, background processing to a worker queue, an to AICTE SSO/e-sign infrastructure.
- Set PRAMAAN_SECRET to a random server-side secret in pro demo passwords, restrict CORS, configure retention, encrypt monitor model/flag precision by data stratum.
- Use shadow mode before decisions are affected. An engine th agreed threshold remains advisory-only.

10 · API and reference links

- Core API: POST /api/login; GET /api/me; GET /api/ai/status; GE /api/applications; POST /api/applications; POST /api/applicatio POST /api/audit/verify.
- OpenRouter API: https://openrouter.ai/docs/api_reference/aut https://openrouter.ai/openrouter/free.
- Official Ollama API: https://docs.ollama.com/api/generate and https://docs.ollama.com/api/authentication.
- Google Gemini keys: https://ai.google.dev/gemini-api/docs/ap restrict a key in your own Google project.
- PaddleOCR: https://paddlepaddle.github.io/PaddleOCR/main/e
- DigiLocker/API Setu: https://apisetu.gov.in/digilocker and https://docs.setu.co/data/digilocker/quickstart.
- Repository: https://github.com/yoursexybhavya/pramaan-ai-si Production demo: https://pramaan-ai-sih26261.vercel.app.